

Value Chain Physics: Industrial Validation and Formal Proof of Open Complex Giant Systems Based on Non-IID Theory

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Abstract

This paper establishes a Meta-Model and topological existence proof for the governance architecture of Open Complex Giant Systems (OCGS). Addressing the computational breakdown and system instability caused by the implicit Independent and Identically Distributed (IID) assumption in traditional Industrial Engineering (IE), Operations Research (OR), and Enterprise Resource Planning (ERP/APS) systems, this paper establishes Non-Identical and Non-Independently Distributed (Non-IID) as the first principle based on Qian Xuesen's OCGS theory. 18 years of real-world industrial practice (IPS system and IPC physics engine) demonstrate that Open Complex Giant Systems possess complete qualitative computability in the physical world, successfully absorbing state space combinatorial explosions of order $\mathcal{O}(N!)$. We resolve apparent theoretical rifts across a five-dimensional self-consistent manifold and construct an empirical defense matrix containing 9 Karl Popper falsifiability protocols.

This paper resolves theoretical rifts across a five-dimensional self-consistent manifold: 1. **Ontological Topology (Field & Boundary Disambiguation)**: Defines top-level cybernetics without a global objective function as the non-existence of an OR-style centralized differentiable cost function, framing global potential $H(\mathbf{X})$ as a non-smooth Lyapunov Barrier Landscape shaping boundaries, while local sub-systems execute gradient sliding on tangent sub-manifolds. 2. **Two-Scale Dynamics (Category Decoupling)**: Eliminates category mistakes between state convergence and structural evolution, establishing micro closed-loop operator $\mathcal{T}_{\text{closed}}^{(S)}$ convergence in Banach space to a unique physical state equilibrium $\mathbf{X}^*$, while second-order meta-cognitive operator Φ drives macroscopic topological jumps to an adaptive dissipative structure attractor S^* under empty feasible domains ($\mathcal{C}(t) = \emptyset$). 3.

Algebraic Geometry (Causal Wavefront Propagation):

Identifies the physical essence of pruning operator Π_{cut} as strictly partially-ordered Causal Wavefront Propagation, leveraging time-unidirectionality to zero out causally unreachable void phase space and losslessly break $\mathcal{O}(N!)$ permutation symmetry redundancy, completing alignment with Classical Computability Theory (P-time Partial Order Propagation). 4.

Phenomenological Dissipation (Onsager Statistical Thermodynamics): Connects KPI vector interference into the Onsager Dissipation Function, formalizing entropy production rate $\dot{S}_{\text{prod}}$ and proving mathematical

isomorphism between physical impedance $\lambda_j = \frac{\partial W_{\text{eff}}}{\partial b_j}$,

Lagrange multipliers, and constraint reaction forces. 5.

Epistemological Inversion (Unbiased Sampling & Defense Matrix): Establishes unbiased sampling protocols for 9 Karl Popper empirical falsifiability deadlines anchored in MES physical warehousing facts, closing an academic defense matrix against OR, management science, and probabilistic AI.

Keywords: Value Chain Physics; Open Complex Giant Systems (OCGS); Non-IID; Qualitative Computability; Lyapunov Barrier Landscape; Onsager Dissipation Function; Causal Wavefront Propagation; Two-Scale Dynamics Decoupling; Unbiased Sampling Protocols; Popper Empirical Falsifiability

0. Methodological Positioning & Qualitative Computability Grounding

Before entering formal mathematical derivations, we establish the epistemological positioning and methodological grounding of this paper: **This paper constructs a Meta-Model and topological existence proof for OCGS governance architectures, where quantitative engineering data serves as empirical validation rather than theoretical boundaries.**

0.1 Why Must it Be “Qualitative”?

Strongly coupled Non-IID Open Complex Giant Systems possess no globally continuous differentiable analytical solutions in full phase space, and micro-parameters drift in real time with dynamic environments. Classical numerical optimization represents “first-order parameter computation,” which inevitably falls into a factorial black hole ($\mathcal{O}(N!)$) in strongly coherent Non-IID giant systems. Any attempt at first-order parameter solving for local quantitative parameters causes macro-divergence due to micro-perturbations. Thus, the sole viable paradigm for governing OCGS is building a **topological geometry and mechanism design Meta-Model** that transcends specific parameters.

0.2 What Does Qualitative Computation Compute?

Qualitative Computability computes system **geometric phase space boundaries, causal topological order (DAG), and micro-state monotonic convergence within tangent spaces**. It answers the topological prerequisite of *how systems become computable*. 18 years of industrial physical practice (IPS system and IPC physics engine, 2004–2022) serve as the sole physical source of qualitative cognition: proving that Open Complex Giant Systems achieve real-world computability, self-healing, generation, survival, and generational evolution. Empirical scale encompassing 50,000 discrete orders and 2,000,000 SKU-site nodes constitutes decisive evidence of absorbing state space factorial combinatorial explosion.

0.3 Alignment with Classical Computability Theory (Church-Turing Thesis)

Under the Church-Turing Thesis and computational complexity theory, Non-Deterministic Turing Machines (NTM) traversing full phase space suffer exponential factorial explosion $\mathcal{O}(N!)$. The causal wavefront pruning operator Π_{cut} established in this paper constrains unordered NTM searches into **deterministic polynomial-time partial order propagation (P-time Partial Order Propagation)**. Topologically, this causal orthogonal pruning proves the existence of a projection from non-convex phase space to complete Banach tangent space, completing rigorous proof of OCGS qualitative computability.

I. Introduction & Physical Foundation

A. Physical Observation: Micro-Vector Interference and Onsager Dissipation

In complex discrete manufacturing, enterprises face systemic traps: high inventory, high stockout, high cost, high delay, and high complexity (“five-highs and two-lows”). Classical management attributes this to “poor execution” or “inaccurate data.” From non-equilibrium thermodynamics and cybernetics, the root cause is the **non-orthogonal collision and interference cancellation of sub-system local KPI vectors**.

This is not a managerial attribution issue. Organizational internal friction in supply chains is mathematically isomorphic to non-equilibrium thermodynamic systems, satisfying the inner product balance structure of flux and forces required by Onsager reciprocal relations. The Onsager dissipation function and generalized thermodynamic forces referenced in this paper represent topological isomorphic mappings of cybernetics and non-equilibrium statistical physics in abstract phase spaces, with both flux vectors and driving forces non-dimensionalized. Formalizing internal friction

via dissipation functions is not a metaphor—it is a structural isomorphism.

When Sales, Production, Procurement, and Finance pursue local optimal KPIs, their response flux vectors $\mathbf{v}_i$ collide at obtuse angles with generalized thermodynamic forces $-\nabla_{\mathbf{x}_i} H$, minimizing the composite vector magnitude:

$$\left\| \sum_{i=1}^M \mathbf{v}_i \right\| \ll \sum_{i=1}^M \|\mathbf{v}_i\|$$

where $\mathbf{v}_i$ denotes the action response flux vector of the i -th subsystem (Sales, Production, Procurement, Finance) and $\|\cdot\|$ represents the vector norm.

Under the **Onsager Dissipation Function** and non-equilibrium thermodynamics, the internal entropy production rate $\dot{S}_{\text{prod}}$ is expressed as:

$$\dot{S}_{\text{prod}} = \sum_{i=1}^M \mathbf{J}_i \cdot \mathbf{X}_i = \sum_{i=1}^M \mathbf{v}_i \cdot (-\nabla_{\mathbf{x}_i} H) \geq 0$$

where $\dot{S}_{\text{prod}}$ represents internal non-equilibrium entropy production rate, $\mathbf{J}_i = \mathbf{v}_i$ denotes thermodynamic response flux of subsystem i , and $\mathbf{X}_i = -\nabla_{\mathbf{x}_i} H$ represents generalized thermodynamic force derived from potential landscape $H(\mathbf{X})$.

Obtuse vector interference generates massive **Vector Cancellation Heat** (ΔW_{heat}). Effective work efficiency approaches zero, and management energy dissipates into organizational static friction:

$$F_{\text{friction}} = \mu_{\text{org}} N_{\text{silos}}$$

and Onsager entropy production. In the equation, $\mathbf{F}_{\text{friction}}$ is static friction force vector, μ_{org} is dimensionless friction coefficient, and N_{silos} represents departmental silo damping magnitude.

B. Core Dilemma: Non-IID Coupling, Factorial Redundancy & Causal Wavefronts

Traditional ERP/MRP/APS software implicitly assumes phase-space elements are Independent and Identically Distributed (IID). However, real discrete manufacturing exhibits strong, time-varying, non-linear Non-IID coupling with non-zero Hessian off-diagonal entries:

$$A_{ij} = \frac{\partial^2 H}{\partial x_i \partial x_j} \neq 0$$

where A_{ij} denotes the cross-coupling entry of the Non-IID coherent tensor between state variables x_i and x_j .

Compared to celestial mechanics and statistical physics, qualitative solving of discrete manufacturing OCGS encounters **three intrinsic boundary constraints**:

1. **Endogenous micro-gaming and subjective goals**: System states embed human behavior and goals, rendering Mean-Field Approximations invalid; 2. **Strong coherent cascade coupling**: Interdependent physical elements invalidate Independent and Identically Distributed (IID) statistical assumptions; 3. **Physical irreversibility & zero-tolerance**: Rigid delivery deadlines demand deterministic precision, excluding probabilistic auto-regressive hallucinations.

Full-degree-of-freedom permutation symmetry redundancy causes state space possibilities to experience combinatorial explosion ($\mathcal{O}(N!)$). Classical OR solvers (MIP/LP) fail under non-convex fractured manifolds because they treat causally impossible permutations as valid possibilities.

From algebraic geometry, $\mathcal{O}(N!)$ factorial complexity is mathematical symmetry redundancy. Physical materials and processes possess natural time unidirectionality and partial order causality. The pruning operator Π_{cut} executes **strictly partially-ordered Causal Wavefront Propagation**, setting causally unreachable void phase space to zero without cutting any physically feasible bases, losslessly projecting non-convex manifolds onto complete Banach tangent spaces with positive-definite Hessians.

C. Transition: From 18-Year Industrial Physical Practice to the Qualitative Axiomatic System

An industrial empirical scale encompassing 50,000 discrete orders and 2,000,000 SKU-site nodes (Lenovo global supply chain IPS system and IPC engine, 2004–2022) thoroughly validates the capacity of this qualitative model to absorb factorial combinatorial explosion ($\mathcal{O}(N!)$) in real non-convex physical environments. Truth in physics and cybernetics is never deduced purely a priori; **18 years of industrial physical practice represent the sole physical source of our Qualitative Axiomatic System**—“no empirical practice, no cognitive insight”.

In this 18-year trajectory, the primary breakthrough is **qualitative**: proving that Open Complex Giant Systems (OCGS) achieve real-world computability, self-healing, generation, survival, and generational evolution: - **50,000**

Discrete Orders & 2,000,000 SKU-Site Nodes:

Demonstrates engine capacity to absorb Non-IID combinatorial state space explosion; - **150,000 Non-Convex Inequality Constraints & Engineering Changes**: Demonstrates manifold collapse by business ontology & single-brain singularity; - **Human-Out-of-the-Loop Autonomous Rate >95%**: Proves silicon-based high-frequency self-healing beyond human bandwidth limits; - **48-Hour Delivery Response Rate Improved from 50%-60% to >98%**: Demonstrates convergence to unique self-consistent state equilibrium after eliminating vector cancellation heat; - **Asset Turnover Velocity Increased by 1.9x**: Proves effective work maximization via Lagrange dual impedance bottleneck focusing.

II. Qian Xuesen's OCGS Theory & Paradigm Shift

A. From Metasynthesis to Complex Giant Systems: The Physical Constitution

In the late 20th century, Qian Xuesen proposed Open Complex Giant Systems (OCGS) theory and Metasynthetic Engineering, envisioning the human-machine “Hall of Workshop for Meta-Synthetic Engineering” and “General Design Department.”

Value Chain Physics provides a formal, engineering realization of Qian's vision. We recognize that global enterprise supply chains are typical Open Complex Giant Systems—exhibiting openness (exchanging matter, energy, and information with market environments), complexity (Non-IID strong coupling), and giant scale (500,000 demand orders and 2,000,000 SKU-site nodes).

B. Harmonization of Qian's “Human-Centric” & “Human-out-of-the-Loop” Cybernetics

Qian Xuesen's theory emphasizes human-machine integration with “human as the core.” Our “human-out-of-the-loop (>95% autonomous rate)” refers specifically to high-frequency L0/L1 routine operational netting, where the silicon engine absorbs combinatorial burdens exceeding human bandwidth (<50 bits/s vs manifold entropy growth $>10^4$ bits/s).

Qian Xuesen's ‘human as the core’ refers to retaining human sovereignty at the qualitative boundary-shaping and ultimate adjudication level; our ‘human-out-of-the-loop’ refers to exiting high-frequency operational execution while remaining present in the second-order constitutional loop. These are not two conflicting views of the same ‘human,’ but two distinct operational levels of human authority: sovereignty belongs to humans, while control is hierarchically decoupled.

Three Cybernetic Levels of “Human-out-of-the-Loop”: Human-out-of-the-loop is hierarchical elevation rather than human exclusion. L0/L1 execution loops are

governed by silicon engines, while L2/L3 constitutional loops are held by human chief architects. Human observers elevate to L2/L3 **second-order meta-cognitive constitutional adaptation and General Design**

Department (Φ operator), defining prior normative boundaries, overriding axiomatic bases, and addressing unknown black swan phase transitions. This fulfills Qian’s vision of the “General Design Department + Hall of Workshop.”

III. The Eight Qualitative Axioms of Value Chain Physics

A. Core Mathematical Notation Table

The table below exhaustively lists core cybernetic variables and physical meanings across the Eight Axioms:

Symbol	Mathematical Definition	Physical & Cybernetic Meaning
$X(t)$	State Vector in Phase Space	High-dimensional physical state vector comprising demand, inventory, WIP, deadlines, and engineering changes.
$H(X)$	Lyapunov Barrier Landscape	Global scalar potential field shaped by conservation laws and physical hard constraints, defining topological feasibility bounds.
v_i	Subsystem Action Flux Vector	Microscopic action response flux vector of subsystem i pursuing local optimal KPIs.
$\dot{S}_{\text{prod}}$	Onsager Entropy Production Rate	Internal non-equilibrium thermodynamic entropy production rate caused by force-flux mismatch.
A_{ij}	Non-IID Coherent Tensor Element	Hessian matrix entry $\frac{\partial^2 H}{\partial x_i \partial x_j} \neq \mathbf{0}$, representing strong inter-element coupling.

Symbol	Mathematical Definition	Physical & Cybernetic Meaning
$\mathcal{D}$	Atomic Data Model (Substance / Body)	Indivisible atomic data model representing physical reality.
$\mathcal{A}$	Atomic Business Algorithm (Function / Use)	Indivisible atomic business algorithm executing physical rules.
$\mathcal{C}$	Single-Brain Singularity	$\mathcal{C} = \text{Entangled} \langle B_{\text{ph}} \rangle$, entangled three-dimensional collapse state integrating physical business modeling, atomic data modeling, and system integrated architectural capability.
Δ	Norm Residual Difference	Double-norm residual difference $\Delta = \ X_{\text{phys}} - X_{\text{plan}} \ $ between execution and plan.
λ_j	Physical Impedance / Lagrange Multiplier	Marginal contribution of relaxing bottleneck constraint b_j to effective work W_{eff} , isomorphic to Lagrange multiplier.
Φ	Second-Order Meta-Cognitive Operator	High-order operator executing constitutional rewriting beyond first-order formal systems.
Π_{cut}	Causal Wavefront Pruning Operator	Orthogonal projection pruning operator zeroing causally unreachable void phase space based on time unidirectionality.

Symbol	Mathematical Definition	Physical & Cybernetic Meaning
X^*	Microscopic Equilibrium State	Unique physical state equilibrium solution under closed-loop evolution operator in fixed structure S .
S^*	Macroscopic Dissipative Attractor	Macroscopic adaptive structure attractor driven by second-order operator Φ during environmental phase transitions.
Q, R	Canonical Quadratic Representation Matrices	Quadratic weighting matrices representing local state and control energy costs on tangent sub-manifolds.
γ	Tangent Space Strong Convexity Parameter	Minimal eigenvalue $\lambda_{\min}(Q) > 0$ guaranteeing local strong convexity.
k	Lipschitz Contraction Factor	Contraction parameter $k = 1 - \frac{\gamma(\ Q\ + \ R\)\Delta t}{2} \in (0,1)$ ensuring Banach fixed-point convergence.

B. Methodological Warning: Avoiding Point, Linear, and MECE Epistemological Collapse

Before examining the Eight Axioms, a methodological warning must be established:

Without a high-dimensional phase space topological map, cognitive understanding collapses into three forms: **Point Collapse** (discrete checklists), **Linear Causality** (erasing high-order coupling and gaming), and **MECE Prisoners** (slicing holistic networks). MECE implicitly assumes element-wise independence (IID); forcing MECE in Non-IID strongly coupled physical networks is akin to slicing a holistic spiderweb, destroying atomic self-consistency. Readers must transcend point, linear, and MECE collapse when reading the Eight Axioms.

C. Dual Governance Mechanism

As the core bridge connecting qualitative axioms to formal mathematical proofs, this paper establishes a **Dual Governance Mechanism** to resolve the apparent

semantic rift between Axiom I (Global Potential Landscape) and “no top-level objective function”:

Top-Level Barrier Shaping Boundaries: Top-level cybernetics refrains from constructing a centralized differentiable cost function, framing global potential $H(X)$ as a **non-smooth Lyapunov Barrier Landscape** shaped by physical conservation laws and hard constraints, defining boundaries via barriers and DAG topologies to strip non-convex fractures and incoherent degrees of freedom. **2. Local Tangent Gradient Sliding:** Within local convex compact sets $\Omega_{\Delta t}^{\text{convex}}$ restricted by topological pruning, local subsystems execute gradient sliding along negative gradients within tangent sub-manifolds.

D. Axiom I: Teleology (Global Potential Landscape Gradient Axiom)

Axiom I (Global Potential Landscape Gradient Axiom): *Open Complex Giant Systems must possess a single self-consistent global scalar potential landscape $H(X)$, where sub-module driving forces are governed by its negative gradient:*

$$F_{\text{global}} = -\nabla H(X)$$

where F_{global} denotes global gradient driving force vector, $H(X)$ is global scalar potential landscape, and X is physical state vector. According to the Dual Governance Mechanism (Section III.C), $H(X)$ is locally smooth within local convex tangent sub-manifolds $\Omega_{\Delta t}^{\text{convex}}$ where $\nabla H(X)$ represents standard negative gradient forces, while acting as a boundary-shaping barrier function at non-smooth barrier boundaries.

[Corollary 1.1: Vector Interference and Onsager Heat Dissipation]

Performance evaluations centered on departmental silos induce vector collisions at obtuse angles under Non-IID coupling. In non-equilibrium thermodynamics, this maximizes internal entropy production $\dot{S}_{\text{prod}}$:

$$\Delta W_{\text{heat}} = \sum_{i=1}^4 \|v_i\| - \left\| \sum_{i=1}^4 v_i \right\| > 0$$

Subsystem optimal collisions minimize composite vector magnitudes, driving effective work to zero. Systems must establish a unified global potential

landscape $H(\mathbf{X})$ to minimize vector cancellation heat $\Delta W_{\text{heat}} \rightarrow 0$.

[Engineering Criteria & Diagnosis]

- **Target:** Departmental KPI systems, siloed local optimization.
- **Diagnosis:** Local potential misalignment induces vector cancellation, driving effective work to zero and maximizing waste heat.
- **Positioning:** Must establish global potential landscape $H(\mathbf{X})$ to set normative boundaries.

E. Axiom II: Ontology (Non-IID Strong Coupling Axiom)

Axiom II (Non-IID Strong Coupling Axiom): Elements within complex manufacturing phase spaces adhere to Non-Identical and Non-Independently Distributed (Non-IID) strong coherent coupling:

$$A_{ij} = \frac{\partial^2 H}{\partial x_i \partial x_j} \neq 0$$

where A_{ij} is the Non-IID coherent tensor element indicating non-zero cross-coupling between state variables x_i and x_j .

[Corollary 2.1: State Space Explosion & Human-Out-of-the-Loop Computation]

Under Non-IID coupling, full-degree-of-freedom permutation symmetry causes state space possibilities to experience factorial explosion $\mathcal{O}(N!)$, far exceeding human cognitive limits (Miller's 7 ± 2 , bandwidth < 50 bits/s). Open-loop scheduling relying on human experience or Excel falls into factorial black holes. Systems must establish "human-out-of-the-loop (>95% autonomous rate)" silicon automation based on rigid tangent space pruning.

[Engineering Criteria & Diagnosis]

- **Target:** Human Excel scheduling, unpruned global MIP/LP solving, Large Language Model (LLM) probabilistic fitting models, AI guardrails.
- **Diagnosis:** Carbon-based compute limits cause human cognitive lockup; LLM/ML represents probabilistic fitting whereas supply chains demand zero error tolerance—a single probabilistic hallucination leads to material mismatch and assembly line shutdown. So-

called "AI guardrails" are merely L1 software patches that cannot replace L0 rigid physical tangent space pruning.

- **Positioning:** Must establish human-out-of-the-loop rigid silicon solvers; LLM/ML serves only for L2 interaction or document summarization.

F. Axiom III: Scheme (Business Ontology Equation Axiom)

Axiom III (Business Ontology Equation Axiom): All computable, executable enterprise business entities strictly adhere to the business ontology equation:

Business Ontology
 $\equiv$ Data Model (Substance, $\mathcal{D}$)
+ Business Algorithm (Function, $\mathcal{A}$)

- **Atomic Data Model ($\mathcal{D}$):** Represents true physical state; - **Atomic Business Algorithm ($\mathcal{A}$):** Executes physical rules and actions.

[Corollary 3.1: Reductionist Dissection Fallacy]

Traditional enterprise architecture separates business logic, data modeling, software engineering, and consulting. This reductionist dissection destroys the atomic self-consistency of data models ($\mathcal{D}$) and algorithms ($\mathcal{A}$), inducing model truncation errors and system oscillations. Systems must maintain atomic self-consistency.

[Engineering Criteria & Diagnosis]

- **Target:** Siloed business, data, IT, and consulting engineering models (e.g., panacea microservices or data mid-office myths).
- **Diagnosis:** Dissected development destroys ontological self-consistency, generating massive truncation errors across interfaces and causing over 85% of APS projects to regress to Excel.
- **Positioning:** Must preserve atomic indivisibility of data models (Substance) and business algorithms (Function).

G. Axiom IV: Capability (Three-Dimensional Entangled State Collapse Axiom)

Axiom IV (Three-Dimensional Entangled State Collapse Axiom): High-dimensional complex manifolds cannot be

constructed via hierarchical committee meetings; they must be collapsed by a Single-Brain Singularity possessing physical business modeling, atomic data modeling, and system integrated architectural capabilities:

$$\mathbf{C} = \text{Entangled}(\mathbf{B}_{\text{phys}}, \mathbf{D}_{\text{topo}}, \mathbf{A}_{\text{arch}})$$

where $\mathbf{C}$ represents the Single-Brain Singularity entangled state, $\mathbf{B}_{\text{phys}}$ denotes physical business modeling, $\mathbf{D}_{\text{topo}}$ is atomic data modeling, and $\mathbf{A}_{\text{arch}}$ is system integrated architectural capability.

[Corollary 4.1: Single-Brain Singularity & Organizational Topology Institutionalization]

Hierarchical multi-head committees or ungoverned multi-Agent free collaborations are equivalent to Nash equilibrium gaming deadlocks, causing communication friction to explode exponentially. Misattributing structural topology dividends to individual execution, or relying on personal experience without institutionalized topological lock-in, inevitably leads to system collapse. Only a “Single-Brain Singularity” can complete high-dimensional constraint collapse and freeze computational mechanisms into institutionalized topology.

[Engineering Criteria & Diagnosis]

- **Target:** Bureaucratic multi-head committees, ungoverned multi-agent free collaboration, organizations lacking institutionalized topological lock-in.
- **Diagnosis:** Communication friction explodes exponentially; lack of institutionalized topological lock-in guarantees system collapse upon key personnel turnover.
- **Positioning:** Single-Brain Singularity collapses high-dimensional constraints; institutionalized structures lock in computational mechanisms.

H. Axiom V: Mechanism (State Residual Self-Healing Closed-Loop Axiom)

Axiom V (State Residual Self-Healing Closed-Loop Axiom): Systems must establish high-frequency “plan-execution-feedback” closed loops driven by double-norm residual differences:

$$\Delta = \|\mathbf{X}_{\text{phys}} - \mathbf{X}_{\text{plan}}\|$$

where Δ represents norm residual difference, $\mathbf{X}_{\text{phys}}$ is real-time physical

execution feedback vector, $\mathbf{X}_{\text{plan}}$ is top-level plan vector, and $\|\cdot\|$ denotes vector norm in high-dimensional space.

[Corollary 5.1: Residual-Driven Closed-Loop Self-Healing]

Systems lacking real-time execution deviation feedback operate under open-loop control, where physical errors inevitably accumulate and cause plan divergence. Static dashboards and open-loop control towers cannot reduce phase space complexity. Only closed-loop feedback driven by norm residual difference Δ enables autonomous self-healing computation without human gaming friction.

[Engineering Criteria & Diagnosis]

- **Target:** Open-loop plan dispatch, ERP/APS lacking real-time feedback mechanisms, static digital twins, supply chain control towers.
- **Diagnosis:** Refreshing UI displays does not reduce underlying complexity; control towers remain post-hoc rear-view mirror observation layers.
- **Positioning:** Residual-driven human-out-of-the-loop self-healing steering mechanism.

I. Axiom VI: Path (Micro-Physical Observability Axiom)

Axiom VI (Micro-Physical Observability Axiom): Global state construction must anchor bottom-up in front-line physical execution facts, following micro-physical observability.

[Corollary 6.1: Bottom-Up Micro-Physical Observability Pull]

Top-down data aggregation suffers exponential distortion due to multi-layered administrative abstraction and beautification. Cybernetic governance must anchor bottom-up in physical execution front-lines (L0/L1 daily execution layer), pulling measurements backward.

[Engineering Criteria & Diagnosis]

- **Target:** Pure top-down S&OP static planning, top-down administrative reporting, static dashboards.
- **Diagnosis:** Ignorance of front-line execution invalidates master planning; dashboards serve as post-hoc rear-view mirrors.
- **Positioning:** Bottom-up physical anchoring in execution facts.

J. Axiom VII: Dynamics (Physical Impedance Dual & Lagrange Isomorphism Axiom)

Axiom VII (Physical Impedance Dual & Lagrange Isomorphism Axiom): *Driving force overcoming organizational friction originates from physical impedance focusing on bottleneck constraints:*

$$\lambda_j = \frac{\partial W_{\text{eff}}}{\partial b_j}$$

where λ_j denotes physical impedance of the j -th bottleneck constraint, W_{eff} represents global effective work, and b_j is the physical capacity upper bound of bottleneck resource j .

Penetration force $\mathbf{F}_{\text{penetration}}$ interferes constructively with physical work vectors to overcome organizational static friction $\mathbf{F}_{\text{friction}} = \mu_{\text{org}} \mathbf{N}_{\text{silos}}$ and break bottlenecks:

$$\mathbf{F}_{\text{penetration}} = \mathbf{F}_{\text{logic}} + \mathbf{F}_{\text{will}}$$

[Isomorphic Connection of Lagrange Multipliers and Constraint Reaction Forces]

Physical bottleneck impedance λ_j is mathematically isomorphic to Lagrange Multipliers (shadow prices) in variational optimization, and constraint reaction forces in classical mechanics. Adhering to Occam's Razor, this paper preserves Newtonian work and impedance as first-principle physical ontology, recognizing Lagrange multipliers as the mathematical projection of physical impedance onto differential phase space.

[Corollary 7.1: Physical Impedance Focusing & Constructive Penetration Force]

Evenly spread administrative reforms are inevitably absorbed by organizational static friction. Inertial management actions lacking physical impedance driving forces collide orthogonally with physical work vectors (interference angle $\theta \rightarrow 90^\circ$), reducing effective work to zero and dissipating as friction heat. Only by leveraging physical impedance λ_j to quantitatively focus on bottlenecks can logic and willpower synthesize into sharp penetration forces that break organizational barriers.

[Engineering Criteria & Diagnosis]

- **Target:** Evenly spread executive mandates, inertial management lacking

impedance focus, non-technical dictation to operations.

- **Diagnosis:** Management vectors collide orthogonally with physical work vectors; effective work drops to zero and converts into static friction heat.
- **Positioning:** Quantitative physical impedance focusing to penetrate organizational resistance.

K. Axiom VIII: Evolutionism (Evolutionism & Two-Scale Dynamics Decoupling Axiom)

Axiom VIII (Evolutionism & Two-Scale Dynamics Decoupling Axiom): *In steady-state operation, system structure is represented by a binary tuple:*

$$S_{\text{static}} = (\mathcal{D}, \mathcal{A})$$

Under Non-IID black swan disruptions and empty feasible domains ($\mathcal{C}(t) = \emptyset$), structure elevates to a ternary tuple:

$$S_{\text{dynamic}} \equiv (\mathcal{D}, \mathcal{A}, \Phi)$$

where S_{static} is static binary tuple, S_{dynamic} is dynamic ternary tuple, $\mathcal{D}$ is data model, $\mathcal{A}$ is business algorithm, and Φ is second-order meta-cognitive operator.

[Corollary 8.1: Category Decoupling of Two-Scale Dynamics (Micro-State Convergence vs Macro-Structural Attractor)]

Based on structural ternary tuple $S = (\mathcal{D}, \mathcal{A}, \Phi)$, system dynamics strictly decouple across two distinct categories: 1. **Microscopic Scale (L0/L1 Physical State Convergence):** Under fixed structure $S = (\mathcal{D}, \mathcal{A})$, closed-loop operator $\mathcal{T}_{\text{closed}}^{(S)}$ forms a strict contraction mapping on complete Banach tangent space, converging exponentially to a unique physical state equilibrium $\mathbf{X}^*$; 2. **Macroscopic Scale (L2/L3 Topological Structural Evolution):** When severe external disruptions break feasible domains ($\mathcal{C}(t) = \emptyset$), second-order meta-cognitive operator Φ steps outside the formal system to rewrite prior axiomatic bases $\mathcal{K}_t \rightarrow \mathcal{K}_{t+1}$, driving topological jumping toward macroscopic adaptive dissipative structure attractor S^* .

[Engineering Criteria & Diagnosis]

- **Target:** Attempting first-order parameter tuning to fix second-order structural

defects; topological loss upon architect transition.

- **Diagnosis:** Failure to decouple first-order execution from second-order constitutional amendment; if autonomous rate drops below 50% within 6 months of chief architect change, the implementation is declared bankrupt.
- **Positioning:** Explicitly decouple first-order work execution from second-order constitutional amendment.

IV. Formal Proof of OCGS Qualitative Computability and Banach Convergence

A. Formal Mathematical Bridge from Axioms to Theorem 1

Theorem 1 serves as the formal mathematical bridge between Axiom III (Scheme) and Axiom V (Mechanism). Axiom III establishes the atomic indivisibility of data models and business algorithms ($\text{Ontology} \equiv \mathcal{D} + \mathcal{A}$), while Axiom V establishes residual-driven closed-loop self-healing ($\Delta = \|\mathbf{X}_{\text{phys}} - \mathbf{X}_{\text{plan}}\|$). Theorem 1 proves that under atomic ontology and real-time residual feedback, the closed-loop operator $\mathcal{T}_{\text{closed}}$ constitutes a strict contraction mapping on the complete Banach tangent space, driving physical state trajectories to converge monotonically to $\mathbf{X}^*$. This is not an isolated addition, but the mathematical projection of the eight axioms.

B. Phase Space Causal Wavefront Propagation & Local Convexification

Connecting to Section III.C Dual Governance Mechanism, non-convex phase space Ξ cannot be solved globally. The pruning operator Π_{cut} executes causal wavefront propagation:

Lemma 1 (Causal Wavefront Pruning & Local Convexification Lemma): Under strict partial order causality, the projection operator Π_{cut} executes Causal Wavefront Propagation, setting causally unreachable void phase space to zero. The projection maps non-convex phase space onto complete Banach tangent sub-manifolds $\Omega_{\Delta t}^{\text{convex}}$ with positive-definite Hessians without cutting any physically feasible bases.

C. Theorem 1 (Closed-Loop State Banach Convergence Theorem) & Category Decoupling

Categorical Pre-Condition & Convergence Decoupling

Before stating Theorem 1, a rigorous categorical distinction must be established: Theorem 1 proves that under a given structure $\mathcal{S} = (\mathcal{D}, \mathcal{A})$, the micro physical state converges to a unique equilibrium $\mathbf{X}^*$. However, $\mathbf{X}^*$ is not the final state of the system—it is the steady-state equilibrium under fixed prior axioms $\mathcal{K}_t$. When external disturbances break the feasible domain ($\mathcal{C}(t) = \emptyset$), the system jumps to a macroscopic adaptive structure attractor $\mathcal{S}^* = (\mathcal{D}, \mathcal{A}, \Phi)$ driven by the second-order operator Φ . These represent convergence across two distinct categories: micro-convergence governs steady-state operation, while macro-jumping governs generational evolution. Theorem 1 guarantees only the former.

By Lemma 1, the local potential function on tangent sub-manifold $\Omega_{\Delta t}^{\text{convex}}$ has a positive-definite Hessian matrix, making it γ -strongly convex ($\gamma = \lambda_{\min}(\mathbf{Q}) > 0$) and L -smooth ($L = \|\mathbf{Q}\| + \|\mathbf{R}\|$).

Theorem 1 (Closed-Loop State Banach Convergence Theorem): Under system structure $\mathcal{S} = (\mathcal{D}, \mathcal{A})$, let $(\Omega_{\Delta t}^{\text{convex}}, \|\cdot\|_{\Delta t})$ be a complete Banach tangent space. If discrete time step Δt satisfies:

$$\Delta t < \frac{2}{\gamma(\|\mathbf{Q}\| + \|\mathbf{R}\|)}$$

then closed-loop evolution operator $\mathcal{T}_{\text{closed}}^{(S)}$ is a strict contraction mapping with Lipschitz factor:

$$k = 1 - \frac{\gamma(\|\mathbf{Q}\| + \|\mathbf{R}\|)\Delta t}{2} \in (0,1)$$

where Δt is discrete step size, γ is tangent space strong convexity parameter, $\mathbf{Q} \succcurlyeq 0$ and $\mathbf{R} \succ 0$ are state and control cost canonical quadratic representations on local tangent spaces measuring topological existence of Lipschitz factor k , without requiring global numerical specification. System physical state trajectories converge exponentially to a unique physical equilibrium $\mathbf{X}^*$:

$$\lim_{m \rightarrow \infty} \left(\mathcal{T}_{\text{closed}}^{(S)} \right)^m (X_0) = X^*$$

V. Karl Popper Empirical Falsifiability Protocols & Academic Defense Matrix

To provide strict Empirical Falsifiability, we establish 9 physical unbiased verification protocols based on physical facts. Should empirical indicators breach designated critical thresholds in ultra-large discrete manufacturing scenarios, the corresponding axiomatic propositions are empirically falsified:

No.	Falsifiable Proposition	Axiom Violated	Unbiased Sampling Protocol	Quantitative Falsification Threshold
F-01	Local KPI optimization converges to global optimum without global potential $H(X)$	Axiom I (Teleology)	Sample MES actual labor hours and flux vectors v_i , computing dissipation ratio $\eta = \Delta W_{\text{heat}} / \sum \ v_i\ $	Spontaneous system collapse when dissipation ratio $\eta > 0.3$
F-02	Human experience / Excel absorbs Non-IID combinatorial explosion	Axiom II (Ontology)	Collect human schedule change rates and plan divergence rates daily for 30 days	Plan diverges and autonomous rate $< 50\%$
F-03	Divided data models and algorithms achieve high-frequency closed-loop self-healing	Axiom III (Scheme)	Sample code interface truncation errors and manual intervention rates	$> 85\%$ modules conflict and regress to manual Excel

No.	Falsifiable Proposition	Axiom Violated	Unbiased Sampling Protocol	Quantitative Falsification Threshold
F-04	Hierarchical committee meetings construct high-dimensional coherent manifolds	Axiom IV (Capability)	Track cross-departmental meeting hours and decision revision frequencies	Communication friction $> \mathcal{O}(1)$ with gaming deadlocks
F-05	Auto self-healing operates without residual norm feedback $\Delta = \ X_{\text{phys}} - X_{\text{plan}}\ $	Axiom V (Mechanism)	Compare physical WMS completion timestamps with plan norm differences Δ	Deviation feedback fails and micro-execution decouples from plan
F-06	Top-down aggregated data remains accurate without bottom-up physical anchoring	Axiom VI (Path)	Compare top-level S&OP data with MES barcode warehousing logs	Top-level data distortion $> 30\%$ causing instruction failure
F-07	Hierarchical resistance is broken without physical impedance dual focusing	Axiom VII (Dynamics)	Measure bottleneck Lagrange multipliers λ_j and reform kinetic decay	Reform resistance drops effective work efficiency $> 50\%$
F-08	Systems evolve under black swans without second-order meta-cognitive operator Φ	Axiom VIII (Evolutionism)	Monitor feasible domain $\mathcal{C}(t)$ under supply chain disruptions	System locks up in deadlock when feasible domain is empty

No.	Falsifiable Proposition	Axiom Violated	Unbiased Sampling Protocol	Quantitative Falsification Threshold
F-09	Response rate < 90% for 3 months, or autonomous rate < 50% within 6 months of chief architect change	Theorem 1 / Axiom VIII	Solely credit MES physical warehousing timestamps against initial MPS promises	Response rate < 90% for 3 months or autonomous rate < 50% after transition

VI. Limitations & Domain Boundaries

1. Physical & Scenario Degradation:

As phase-space coupling weakens to Independent and Identically Distributed (IID) conditions, high-dimensional tangent pruning constraints automatically relax into traditional OR (LP/MIP) and ERP/APS logic.

2. Entity & Game Degradation:

The 18-year validation covers multi-plant coordination under single corporate entities; multi-party games across distinct legal entities require second-order mechanism design.

3. Mathematical & Operator

Degradation: Banach fixed-point convergence k assumes local tangent convexification; under non-continuous phase transitions, carbon-based second-order operator Φ must execute constitutional rewriting.

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